



KARNATAKA STATE POLICE | SCRB

DATATHON 2025

PROBLEM STATEMENT

Intelligent Conversational AI & Crime Analytics Platform

Empowering Investigators, Analysts & Policymakers
with AI-Driven Crime Intelligence

Track: AI/ML

Domain: Law Enforcement

Level: Open

OVERVIEW & BACKGROUND

Challenge Overview

The Karnataka State Police (KSP) and the State Crime Records Bureau (SCRB) maintain extensive crime records capturing incidents, offenders, and victims across the state. However, the current analytical ecosystem faces significant structural hurdles that prevent effective intelligence-led policing. This Datathon challenges participants to design and develop an **Intelligent Conversational AI and Crime Analytics Platform** that enables investigators, analysts, and policymakers to interact with the state crime database using natural language queries while providing advanced analytical capabilities grounded in criminology and sociological insights.

The Challenge: Moving Beyond Manual Records

Data Silos & Manual Processes:	Crime records are managed in independent silos, heavily reliant on Excel-based reporting rather than integrated, automated systems.
Lack of Advanced Analytics:	There is a notable absence of AI-driven approaches, leaving deeper behavioural patterns, social interactions, and interconnected criminal networks undiscovered.
Information Gaps:	The SCRB currently receives limited, fragmented information, hindering its ability to perform comprehensive state-wide analysis.
Reactive vs. Proactive Policing:	Without systematic exploration of emerging trends, investigators lack tools for proactive policing strategies and evidence-based prevention.

What Your Solution Must Enable

Crime Pattern Discovery	Criminal Network Analysis	Socio-Demographic Insights	Behavioural Profiling	Proactive Crime Prevention
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PROPOSED SOLUTION FRAMEWORK — PART I

1 Conversational Crime Intelligence Interface

A natural language chatbot that bridges investigators and complex crime databases without requiring SQL or technical expertise.

- Natural language interface for querying FIRs, accused, victims, locations, and investigation status in **English and Kannada**.
- Context-aware conversations allowing seamless follow-up queries without repeating context.
- Save full conversation history as **PDF locally** for audit and reference.
- Voice interaction support for hands-free Q&A during field operations.
- Multi-turn dialogue management with session memory and intent resolution.

2 Criminal Network & Relationship Analysis

Replace independent data silos with a connected intelligence graph that reveals organized crime structures invisible in isolated records.

- Identification of links between accused, victims, locations, financial accounts, and crime incidents.
- Node-based **network visualization** showing connections between suspects, victims, and recurring locations.
- Detection of organized crime groups, co-offender networks, and repeat offender clusters.
- Repeat offender tracking linking individuals to multiple incidents across jurisdictions with MO analysis.
- Association detection revealing hidden criminal ties impossible to spot in siloed spreadsheets.

3 Crime Pattern & Trend Analytics

Move from reactive reporting to strategic intelligence through statistical and geospatial pattern recognition.

- Analysis of crime trends across time, geography, crime type, and modus operandi.
- District-level drill-down interactive maps to visualize crime patterns across police stations.
- Spatiotemporal cluster detection: 'Crime Hotspot' identification by layering time-of-day with location.
- Emerging trend alerts — visual red-zone indicators when a crime category spikes vs. historical averages.
- Seasonal and event-based crime trend analysis for proactive resource deployment.

4 Sociological Crime Insights

Understand the 'why' behind the 'where' by correlating crime data with socio-economic and demographic indicators.

- Analysis of crime patterns based on age, gender, and socio-economic background.
- Identification of social risk factors influencing crime patterns at district and taluk levels.
- Correlation of crime with urbanization, migration, economic stress, and education indicators.
- Socio-economic overlay: crime data layered with population distribution and urbanization patterns.
- Predictive risk-scoring charts forecasting high-risk areas based on hidden socio-economic correlations.

5 Criminology-Based Offender Profiling

Build data-driven profiles that help investigators prioritize cases and anticipate future offences.

- Identification of repeat offenders and habitual criminals with cross-jurisdiction linkages.
- Behavioural analysis of offenders based on crime history and evolving modus operandi.
- Risk scoring of offenders to prioritize investigation effort and resource allocation.
- Anomaly detection: visual call-outs for incidents deviating from standard behavioural baselines.

- Profile dashboards linking an individual to all related incidents, locations, and associates.

PROPOSED SOLUTION FRAMEWORK — PART II

6 Investigator Decision Support

Accelerate case resolution by surfacing relevant precedents, automating timelines, and recommending investigative leads.

- Automated case summaries and investigation timelines generated from structured FIR data.
- Identification of similar past cases and their investigation outcomes to guide current enquiries.
- AI-driven recommendations for potential investigative leads based on pattern matching.
- Evidence trail visualization showing reasoning paths and correlations used in analysis.
- Compliance with law enforcement accountability requirements through explainable AI outputs.

7 Financial Crime & Transaction Link Analysis

Follow the money — detect financial flows linked to criminal activities and map money-laundering networks.

- Detection of financial transactions linked to criminal activities and suspicious account clusters.
- Identification of money trails and suspicious transaction networks across financial entities.
- Integration with financial crime investigation workflows and hawala transaction mapping.
- Visual network graphs of transaction flows with flagged anomalies and risk scores.
- Cross-referencing accused records with financial account holders for identity linkage.

8 Crime Forecasting & Early Warning

Shift from reactive response to proactive anticipation using AI-driven predictive models.

- AI-driven identification of emerging crime patterns from historical trend data.
- Early warning alerts for repeat crimes, gang activity, or organized crime re-emergence.
- Predictive analysis of potential crime hotspots using spatiotemporal ML models.
- AI/ML-driven detection of hidden correlations, anomalies, and emerging risk typologies.
- Dashboard indicators with configurable alert thresholds for supervisors and policymakers.

9 Explainable AI & Transparent Analytics

Every AI output must be traceable, auditable, and legally defensible for use in law enforcement contexts.

- All AI responses supported by clear data references, evidence trails, and source citations.
- Visualization of reasoning paths, feature weights, and correlations used in analysis.
- Compliance with law enforcement accountability and judicial admissibility requirements.
- Audit logs capturing every query, analytical step, and user action for governance.
- Confidence scores and uncertainty indicators displayed alongside all predictions.

10 Secure Role-Based Access & Governance

Protect sensitive crime data while enabling the right people to access the right information at the right time.

- Role-based access control for investigators, analysts, supervisors, and policymakers.
- Secure handling of sensitive data with encrypted storage and transmission.
- Comprehensive audit logs with full traceability of data access and analytical actions.
- Compliance with data protection frameworks and law enforcement governance standards.
- Multi-factor authentication and session management for high-security operational environments.

ADVANCED VISUALIZATION CAPABILITIES

Participants are expected to move beyond static charts into **dynamic, spatial, and relational storytelling**. The platform must replace Excel-based reporting with interactive, intelligence-grade visualizations.

Intelligence Dashboards & Geospatial Maps

- **District-Level Drill-Down:** Interactive maps allowing SCRB to visualize crime patterns across districts and individual police stations.
- **Spatiotemporal Clusters:** Crime hotspot identification by layering time-of-day with precise location data, enabling proactive resource deployment.
- **Emerging Trend Alerts:** Red-zone pulsing indicators when a specific crime category spikes in a region compared to historical averages.
- **Heatmaps & Choropleth:** Density-based visualizations of crime concentration across administrative boundaries.

Criminological Network & Link Analysis

- **Relationship Mapping:** Node-based visualization connecting suspects, victims, locations, and financial accounts in an interactive graph.
- **Repeat Offender Profiles:** Visual dashboards linking an individual to multiple incidents, highlighting MO evolution across jurisdictions.
- **Association Detection:** Hidden criminal association discovery impossible to surface from isolated Excel records.
- **Timeline Reconstruction:** Chronological event graphs showing the evolution of criminal activity and network growth.

AI-Driven Predictive Dashboards

- **Socio-Economic Correlation:** Crime data overlaid with urbanization patterns, population distribution, and socio-economic indicators.
- **Predictive Risk Scoring:** AI-driven charts forecasting high-risk areas or emerging crime typologies based on hidden correlations.
- **Anomaly Detection:** Visual callouts for incidents deviating from standard behavioural baselines for complex case linkage.
- **Forecast Horizon:** Configurable prediction windows (7-day, 30-day, seasonal) for operational planning.

Technical Expectations

Category	Expected Capabilities
NLP / Conversational AI	Multi-lingual NLU (English + Kannada), intent recognition, entity extraction, context management, voice ASR/TTS
Machine Learning	Supervised classification, unsupervised clustering, anomaly detection, time-series forecasting, graph ML
Data Engineering	ETL pipelines for structured crime data, data quality management, schema harmonization across source systems
Visualization	Interactive geospatial maps, force-directed network graphs, temporal heatmaps, dashboard with drill-down
Security & Governance	RBAC, end-to-end encryption, audit logging, MFA, data masking for sensitive PII fields
Explainability	SHAP/LIME feature attribution, confidence intervals, reasoning trace visualization, evidence citation

EVALUATION CRITERIA

Innovation & AI Depth	25%	Novelty of AI/ML approach, sophistication of NLP, quality of predictive models
Analytical Accuracy	20%	Correctness of crime pattern detection, network analysis precision, forecast quality
Visualization Quality	20%	Interactivity, clarity, geospatial storytelling, dashboard usability for non-technical users
Sociological & Criminological Depth	15%	Depth of socio-economic correlation, offender profiling rigour, criminological validity
Scalability & Governance	10%	RBAC implementation, audit trail quality, data security, compliance readiness
Presentation & Documentation	10%	Clarity of solution narrative, explainability of AI, quality of technical documentation

SUBMISSION DELIVERABLES

D1	Working Prototype / Demo A functional demonstration of the platform showcasing at minimum the conversational interface, one visualization capability, and one analytical module.
D2	Source Code Repository Well-structured, commented codebase with README, setup instructions, and architecture overview hosted on GitHub or equivalent.
D3	Technical Documentation System architecture diagram, data flow, AI/ML model documentation, API contracts, security design, and deployment guide.
D4	Presentation Deck 10–15 slides covering problem understanding, solution approach, demo walkthrough, innovation highlights, scalability plan, and future roadmap.
D5	Dataset Handling Note Description of synthetic/anonymized dataset used for demonstration, data schema adopted, and approach to handling real PII in production deployment.

This problem statement is issued by the Karnataka State Police / SCRB exclusively for Datathon 2025 participants. All solutions developed remain subject to KSP's intellectual property and data governance policies. Participants must use only synthetic or provided anonymized datasets; use of real PII without explicit authorization is strictly prohibited. For queries, contact the organizing committee through official Datathon channels.